

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States mean = 40 hours	AO1.2	B1	Mean = $\frac{1}{\lambda} = 40$ hours
(b)	Obtains correct probability	AO1.1b	B1	$P(\text{time} < 12) = 1 - e^{-12 \times 0.025}$ $= 1 - e^{-0.3} = 0.259$
(c)	Uses 'no memory' property PI	AO3.4	M1	Exponential distribution has no memory
	Obtains probability	AO1.1b	A1	$P(\text{time} > 30) = e^{-30 \times 0.025} = e^{-0.75}$ $= 0.472$
(d)	States or uses new mean (or uses $e^{-0.3}$)	AO3.4	B1	4 consecutive shifts gives $4 \times 12 = 48$ hours
	Finds probability $P(\text{time} > 48)$ (or uses $(e^{-0.3})^4$)	AO1.1a	M1	$P(\text{time} > 48) = e^{-48 \times 0.025} = e^{-1.2}$
	Obtains correct probability	AO1.1b	A1	$= 0.301$
(e) (i)	States Poisson for model of situation given	AO3.3	B1	Poisson identified
	States value for λ (= 0.025 per hour) for model	AO3.3	B1	Po(0.025) per hour
(ii)	Uses $60 \times$ 'their' λ for 'their' Po model Or Uses exponential model to find $P(\text{Time} > 60)$	AO3.4	M1F	For 60 hours of process $\lambda = 60 \times 0.025 = 1.5$
	Obtains correct probability using Poisson or exponential model	AO1.1b	A1	$P(X=0) = \frac{1.5^0 \times e^{-1.5}}{0!} = e^{-1.5} =$ 0.223
				Total 11 marks

Q2.

- (a) Shape of histogram similar to pdf of exponential distribution.

E1

Mean and SD approximately equal.

E1

2

(b) (i) $E(T) = \frac{1}{0.3} = 3.33$
awrt

B1

1

(ii) $P(T \leq 1) = F(1)$

M1

$= 1 - e^{-0.3}$
 $= 0.259$
 awrt

A1

2

(iii) $P(T > 1.75 | T > 1)$
identifies correct probability.

M1

$= \frac{1 - F(1.75)}{1 - F(1)} \left(= \frac{1 - F(1.75)}{0.741} \right)$
 numerator correct

a1

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$= \frac{e^{-0.525}}{e^{-0.3}} = 0.799$

f.t. on answer to (b)(ii)
 B1 for $P(T < 1.75 | T > 1)$ correctly evaluated.

A1f.t.

3

(iv) Let median value be m
 $F(m) = 0.5$
 $1 - e^{-0.3m} = 0.5$

M1

$e^{-0.3m} = 0.5$
 $-0.3m = \ln(0.5)$
 valid attempt to solve

m1

$$m = 2.31$$

cao

A1

Median time interval = 2.31 minutes

3

[11]

Q3.

(a) (i) $f(t) = \frac{1}{10} e^{-\frac{t}{10}} \quad t > 0$

CAO

B1

$f(t) = 0 \quad t < 0$

CAO 0 & both ranges

B1

2

(ii) $F(t) = \int_0^t \frac{1}{10} e^{-\frac{x}{10}} dx =$

*use of $\int f(x) dx$
condone use of t*

M1

$$\left[-e^{-\frac{x}{10}} \right]_0^t = 1 - e^{-\frac{t}{10}}$$

*integral & limits
CAO*

A1A1

3

(b) (i) Required $P(T < 15) = F(15) =$

use of $F(15)$, or integration

M1

0.776 to 0.777

AWFW

A1

2

(ii) Require $P(10 < T < 20) = F(20) - F(10) =$

use of difference of F s or integration

M1

$$0.86466 - 0.63212 = 0.232 \text{ to } 0.233$$

AWFW

A1

2

[9]

Q4.

(a) $f(x) = \frac{1}{10} e^{-\frac{x}{10}}$

B1

For $x > 0$,

$$F(x) = \frac{1}{10} \int_0^x e^{-\frac{t}{10}} dt = \left[-e^{-\frac{t}{10}} \right]_0^x$$

M1A1 A1

4

= AG

(b) $P(X > 15) = e^{-\frac{15}{10}} = 0.223$

M1A1

2

(c) $P(X < 20) | X > 15 = \frac{P(15 < X < 20)}{P(X > 15)}$

M1

$$= \frac{e^{-\frac{15}{10}} - e^{-\frac{20}{10}}}{e^{-\frac{15}{10}}}$$

A1

$$= 0.393$$

Accept 0.394

A1

3

[9]

Q5.

(a) (i) $f(t) = \frac{1}{800} e^{-\frac{t}{800}}$
 CAO

B1

$$f(t) = 0 \quad \begin{array}{l} \text{for } t \geq 0 \\ \text{for } t < 0 \end{array}$$

CAO both

B1

2

(ii) $\sigma = 800$
 CAO

B1

1

(iii) $P(T > +1000) = \int_{1000}^{\infty} \frac{1}{800} e^{-\frac{t}{800}} dt$
 Use of $\int f(t) dt$

M1

$$= \left[-e^{-\frac{t}{800}} \right]_{1000}^{\infty}$$

integral & limits

A1

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$$= e^{-\frac{1000}{800}} = 0.286 \text{ to } 0.287$$

AWFW

A1

3

(iv) $P(T > +1000 \mid T > 1000) =$
 Use of $P(A \& B) / P(A)$

M1

$$\frac{P(T > 2000)}{P(T > 1000)} =$$

OE

A1

$$\frac{\left[-e^{-\frac{t}{800}} \right]_{2000}^{\infty}}{\left[-e^{-\frac{t}{800}} \right]_{1000}^{\infty}} = 0.286 \text{ to } 0.287$$

AWRT

If answer (iv) = answer (iii), then 3 marks

if answer (iii) correct, otherwise 2 marks

A1

3

- (b) Number of components working after 1000 hours ~ binomial

M1

with $n = 4$ and $p = 0.286/7$

f.t. on (iii)

A1f.t.

$$P(X \geq 2) = P(X = 2, 3, 4) =$$

use of either

M1

$$1 - P(X = 0, 1) =$$

$$0.323 \text{ to } 0.326$$

AWFW

A1

4

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Q6.

$$T \sim E(2)$$

- (a) 1

cao; accept 'unity'

B1

1

- (b) $P(S > 5) = P(T > 4)$

4 cao

B1

$$= 1 - \left(1 - e^{-\frac{4}{2}} \right) = e^{-2}$$

use of exponential cdf or pdf

with $\lambda = 0.5$ or 2

M1

$$= 0.135$$

awrt

A1

3

- (c) $P(S < 5 \mid S > 3) = P(T < 4 \mid T > 2)$
use of conditional probability

M1

Exponential has 'no memory' so
use of; may be implied

M1

$$= P(T < 2)$$

2 cao; (even from 5 – 3)

A1

$$= 1 - e^{-1} = 0.632$$

awrt

A1

4

- (d) Probability = $(b)^5 = 0.000044$ to 0.000046
awfw; ft on (b)

B1ft

Implies an extremely rare event
rare event, or equivalent

E1

2

so casts doubt on model

ag

[10]

Examiner reports

Q2.

Many candidates answered part (a) clearly and concisely and scored full marks. Others ignored the request to use the given summary information and simply described the properties of an exponential distribution; this did not gain any credit.

Part (b) was generally well answered except for part (ii) where few candidates were able to identify the required conditional probability correctly. A small minority of candidates treated the given cumulative distribution function as a probability density function and made little progress.

Q3.

This question was well answered. Candidates remembered to state the range ($t \geq 0$) for

which the formula $f(t) = \frac{1}{10} e^{-\frac{t}{10}}$ was valid and also to state that $f(t) = 0$ for $t < 0$. In part (b) candidates favoured the method of integration, rather than using the distribution function $F(t)$, as a probability formula for $P(T \leq t)$.

Q5.

For some candidates this was not a good first question but this certainly did not appear to affect their subsequent attempts at other questions. In part (a)(i), many candidates omitted the range $t < 0$ and/ or 0 for $t < 0$ and, in part (a)(ii), some candidates quoted 800^{-1} for the standard deviation. Nevertheless, this did not prevent them, and indeed almost all candidates, from scoring the 3 marks available in part (a)(iii). In part (b)(iv), only the most able candidates appeared to be aware that, for an exponential distribution, $P(T > t_2 | T > t_1) = P(T > t_2)$ but some other candidates were able to make significant progress, even if not with complete success, by the application of $P(A | B) = P(A \cap B) \div P(B)$. Whilst many candidates realised that part (b) required their answer to part (a)(ii), only the most able candidates also realised that a binomial distribution was then needed to evaluate $P(X = 2, 3, 4)$ or $1 - P(X = 0, 1)$.

Q6.

Answers to this question were generally very poor. At least 50% of candidates used 0.5 for λ rather than 2 and so worked with e^{-2t} . Although these, and most other, candidates were able to answer part (a) correctly, they made somewhat of a mess of part (b) often compounding the error by not changing 5 to 4. Given these errors, correct answers to part (c) were extremely rare with the vast majority of candidates simply attempting $P(3 < T < 5)$.

In part (d), most candidates made some general statement about trains in stations and failed to realise that reference was expected to part (b) and the value of $(P(T > 4))^5$.